Ian Vandevert

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Website

Summary

Data scientist and software engineer completing a PhD in seismology (defending June 2026), specializing in building analysis pipelines for large scientific datasets (∼775,000 events and ∼12 million records). Experience spans the full stack of scientific computing: ingesting and cleaning messy real-world data, designing efficient numerical algorithms, and building analysis and visualization workflows. Comfortable working across Python (NumPy, pandas, SciPy, ObsPy, GeoPandas), high-performance computing environments, Linux, and real-time data streams. Author of peer-reviewed publications and open-source scientific tools.

Education

IGPP, Scripps Institution of Oceanography, UC San Diego — San Diego, California, USA
Ph.D. in Seismology — Profs. Peter M. Shearer and Wenyuan Fan — *2021–2026 (defending June)*

- Dissertation: "Spatial Variations in Earthquake Source Properties"

University of California, Santa Barbara — Santa Barbara, California, USA
B.S. in Earth Science (Geophysics emphasis) — *2016–2019*

Technical Skills

Languages: Python (primary), MATLAB, Fortran
Python Libraries: NumPy, pandas, SciPy, Matplotlib
Geoscience & Geospatial: ObsPy, PyGMT, GeoPandas
Data Engineering: Concurrent and parallel processing, pipelines for large datasets (10M+ records)
Environment: Linux, remote workflows
Methods: Spectral analysis, time-series analysis, signal processing, spatial statistics, inversion methods

Software

py-nfi (github.com/ivandevert/py-nfi) — 2025 – present
End-to-end pipeline for computing the normalized frequency index (nFI) across large seismic catalogs Python, ObsPy, NumPy, pandas, PyGMT, GeoPandas

- Used to process 30 years of California seismicity: ∼775,000 events and ∼12 million seismograms, from raw waveform download through source-property estimation
- Metadata and waveform ingestion, instrument response correction, and spectral computation.

seisplot (github.com/ivandevert/seisplot) — 2022
Interactive seismogram visualization and analysis tool — *Python, ObsPy*

- GUI application for exploring, filtering, and quality-controlling seismogram data, with built-in record section, PSD, spectrogram, and filter response plotting
- Plugin architecture enabling extensible analysis workflows (e.g., phase arrival overlays, seismogram stacking) without modifying the core codebase

sdpy (github.com/ivandevert/sdpy) — 2024
Teaching materials for spectral decomposition workshop at the SCEC Annual Meeting

- Pipeline for waveform download, spectral computation, and spectral decomposition for source parameters
- Designed as a hands-on starting point for workshop attendees learning spectral decomposition methods

Experience

Graduate Student Researcher — 2021 – Present
Scripps Institution of Oceanography, UC San Diego — *San Diego, California, USA*

- Developed methods for characterizing earthquake source properties, applied to catalogs of ∼775,000 events and ∼12 million seismograms
- Introduced the normalized frequency index (nFI), a metric distinguishing spatial variations in earthquake source physics; released as an open-source end-to-end Python pipeline (see `py-nfi` under Software)
- Contributed to the SCEC/USGS Community Stress Drop Validation Study, a multi-institution comparison of stress-drop estimation methods applied to the 2019 Ridgecrest sequence

- Participated in the SCEC Array of Arrays field deployment (2024), assisting in the installation of ∼400 temporary geophones across five Southern California sites (poster)
- 4 peer-reviewed publications and 10 conference presentations (AGU, SCEC, SSA, EGU); see Publications and Presentations

Geotechnical Associate 2019 – 2020
C2Earth, Inc. *Campbell, California, USA*

- Built an internal project-tracking web application (custom HTML/JS frontend, MySQL backend) deployed on the company's local network; replaced a single overloaded spreadsheet of thousands of projects with searchable, editable records adopted by the firm
- Supported residential and commercial construction projects through field and laboratory work: conducted soil compaction and soil analysis tests, and generated settlement contour maps from hydrostatic level surveys of existing buildings

Undergraduate Research Assistant 2018 – 2019
University of California, Santa Barbara *Santa Barbara, California, USA*

- Worked with Prof. Toshiro Tanimoto on cross-correlation of seismic noise in the Los Angeles basin and presented results at the SCEC Annual Meeting (2018)
- Worked with Prof. Robin Matoza on elastic wave propagation simulation

Publications

1. **Vandevert, I.**, Shearer, P.M., Fan, W. "Distinguishing Spatial Variations in California Earthquake Dynamics Using a High- to Low-Frequency Spectral Ratio"; in progress.
2. **Vandevert, I.**, Shearer, P.M., Fan, W. "Using a High- to Low-Frequency Spectral Ratio to Distinguish Variations in Earthquake Source Properties" *BSSA*; accepted.
3. **Vandevert, I.**, Shearer, P.M., Fan, W. "Ridgecrest Aftershock Stress Drops from P- and S-Wave Spectral Decomposition" *BSSA*, 2024. [doi.org/10.1785/0120240133]

Non-first author: Contributing author on two more publications. See ianvandevert.com.

Presentations

1. 2025 AGU, New Orleans. "Distinguishing Spatial Variations in California Earthquake Dynamics Using a High- to Low-Frequency Spectral Ratio." Oral.
2. 2023 AGU, San Francisco. "Estimating Earthquake Source Parameters from S-Wave Maximum Amplitudes: Application to the 2019 Ridgecrest Sequence." Oral.

Posters: 10 posters presented at AGU, SCEC, SSA, and EGU annual meetings (2018, 2022–2026). Full list available at ianvandevert.com or on request.

Teaching Experience

Teaching Assistant Spring–Fall 2025
UC San Diego *San Diego, CA*

- Courses: *Natural Disasters* (Fall 2025), *Geology of the National Parks* (Spring 2025)
- Led weekly discussion sections, three local field trips, office hours, and grading

Workshop Presenter September 2024
SCEC Annual Meeting *Palm Springs, CA*

- Presented a half-hour tutorial on computing stress drop using spectral decomposition, with accompanying open-source Python materials (`sdpy`)

Awards & Service

- **Outstanding Student Presentation Award**, American Geophysical Union (AGU), 2024. *Seismology section; awarded to the top student presenters.*
- **Peer Reviewer:** *The Seismic Record* (2024); *BSSA* (2024–2025).